

State and Local Government Expenditures

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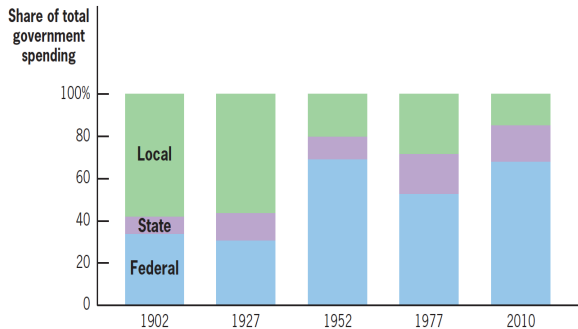
Poll (No Right Answer)

Public schools should be funded primarily via local property taxes.

- A. True
- B. False
- C. Uncertain

- Optimal fiscal federalism: The question of which activities should take place at which level of government
- The distribution of government spending has changed dramatically over time in the United States
- Local state and spending have declined considerably.
- Much state and local spending now supported by intergovernmental grants [transfers from the federal government]

10.1

State and Local Spending in the United States,
1902–2010

Spending and Revenue of State and Local Governments

- Property tax: The tax on land and any buildings on it, such as commercial businesses or residential homes.
- Main source of revenue from local governments due to:
 - ① History: real estate property is visible and hence taxable even in archaic economies with informal businesses
 - ② Immobile tax base: the real estate tax base cannot flee to another jurisdiction (mobility of the tax base is an issue for local governments)
- Today, property tax is about $\frac{1}{3}$ of revenue raised by state+local government (rest is $\frac{1}{3}$ income tax, $\frac{1}{3}$ sales taxes)

The Tiebout (1956) Model

- What is it about the private market that guarantees optimal provision of private goods that is missing in the case of public goods?
- Tiebout's insight was that a factor missing from the market for public goods were shopping and competition
- The situation is different when public goods are provided at the local level by cities and towns
- Competition will naturally arise because individuals can *vote with their feet*: if they don't like the level or quality of public goods provision in one town, they can move to the next town
- This threat of exit can induce efficiency in local public goods production

The Tiebout Formal Model

- We consider a very simple model to illustrate Tiebout's insight and theorem
- Suppose there are $2 \cdot N$ families with identical income Y and 2 towns with N homes each
- Towns 1 and 2 supply level G_1, G_2 of local public schools
- There are 2 types of families:
 - ① N families with kids, with utility $U^K(C, G)$, value private consumption C and schools G
 - ② N elderly families, with utility $U^E(C)$, value only private consumption C

The Tiebout Equilibrium Definition

- Allocation of families across towns is a Tiebout Equilibrium if and only if:
 - ① In each town, G is decided by median voter and financed equally by town residents with budget $Y = G/N + C$
 - ② No two families want to exchange locations across towns
- If majority in town is elderly then $G = 0$ as this maximizes $U^E(Y - G/N)$
- If majority in town is families with kids then $G = G^*$ that maximizes $U^K(Y - G/N, G)$

The Tiebout Theorem

- Tiebout Theorem Part I: In equilibrium, families will sort themselves in towns according to their taste for public good (1 town with elderly only, 1 town with families with kids only)
- Proof: Suppose elderly dominate in town 1 and $G_1 = 0$, then families with kids dominate in town 2 and $G_2 = G^*$. If there is a family with kids in town 1, then there is an elderly family in town 2 and they are willing to switch not an equilibrium.
- Tiebout Theorem Part II: In each town, the level of local public good is efficient
- Proof: In elderly town, $G = 0$ which is efficient as nobody values G .
- In kids town, G^* maximizes $U^K(Y - G/N, G)$ which is also efficient as it is the preferred choice of everybody.

- Tiebout efficiency result holds when there is no cost advantage to everyone living in one big city
- Pure public goods always have a cost advantage (economies of scale). Ex. need only one missile defense shield for country
- Tiebout shows that efficiency can be restored when public good is rival (i.e. an impure public good). Ex. my kid's desk at school displaces another kid from using the desk
- Nevertheless proceed calling the locally publicly provided item a “public good” even though it is not a pure public good

Poll (No Right Answer)

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The Tiebout Model

- People can vote with their feet by choosing the locality that best fits their tastes and provides the best public goods given the tax
- The main message of the model is that competition across local jurisdictions puts competitive pressure on the provision of local public goods:
 - ① Public goods need to reflect tastes of local residents
 - ② Public goods need to be efficiently provided (without waste)

Centralized vs. Decentralized Government

- Conservatives/libertarian tend to like decentralized governments over centralized governments
- Conservatives/libertarian dislike redistribution and like individual choice and competition.
In Tiebout model:
 - ① Local governments do not do any redistribution: individuals receive in local public goods exactly what they are paying in taxes (= benefit principle of taxation)
 - ② Individuals can choose (through their location choice) their preferred mix of public goods and taxes
 - ③ Competition between local govts forces them to provide local public good efficiently

Problems with the Tiebout Model

- The Tiebout model is an idealized model that requires a number of assumptions that may not hold perfectly in reality:
 - ① Individuals can move without any cost across towns
 - ② Individuals have perfect information on the benefits and taxes paid in each town
 - ③ There must be enough towns so that individuals can sort themselves into groups with similar preferences for public goods
 - ④ No externalities/spillovers of public goods across towns [with spillovers across towns, public goods will be under provided in Tiebout model, e.g. parks, police]
 - ⑤ Local govts can charge “poll” taxes (equal payments per person) to residents. In reality, local taxes are property taxes: a flat percentage of home value. Zoning prevents free-riding, e.g. each house must sit on a parcel of at least 6000 sq feet

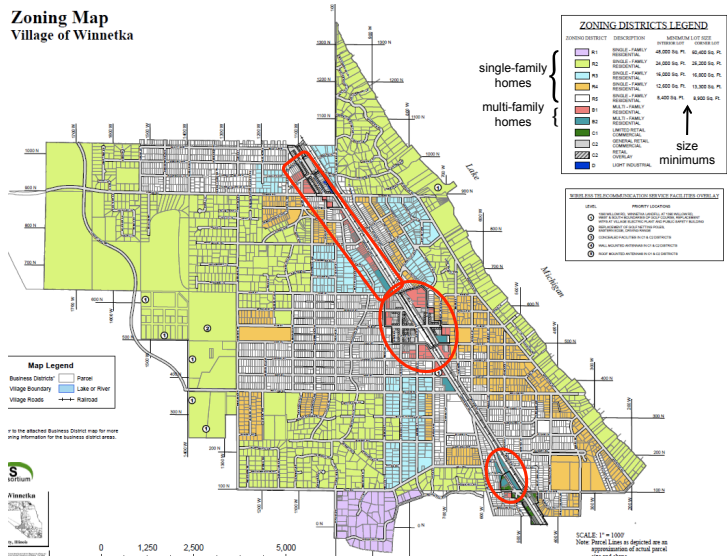
1990 Poll Tax Riots: 200,000 People in Central London



1990 Poll Tax Riots: 200,000 People in Central London



Zoning Example: Winnetka IL



Evidence on the Tiebout Model

- Tiebout Sorting: Resident Similarity Across Areas
- A testable implication of the Tiebout model is that when people have more choice of local community, the tastes for public goods will be more similar among residents than when people do not have many choices
- This evidence is indeed pretty well established
- More Efficiency when there is more Tiebout sorting
- This evidence is controversial

Evidence on the Tiebout Model: Hoxby (2000)

- Hoxby (2000) considers public school districts in the US. She compares cities where:
 - Ⓐ There are few large school districts and hence little choice for residents (such as Miami or LA)
 - Ⓑ There are many small school districts and hence a lot of choice for residents (such as Boston)
- 2 key findings:
 - ① Cities with few districts have less sorting across neighborhood (in terms of school quality) than cities with many districts (this result is well established)
 - ② Cities with many districts have higher test scores on average: this result is controversial (see Rothstein, 2007 critique)

Capitalization of Fiscal Differences into House Prices

- House price capitalization: Incorporation into the price of a house of the costs (including local property taxes) and benefits (including local public goods) of living in the house.
 - High property taxes (relative to public goods quality) depresses housing prices
 - Low property taxes (relative to public goods quality) increases housing prices
- Example: Suppose \$1 cut in property tax (in perpetuity) with no change in value of local public good
 - Capitalized value:
$$\Delta V = 1 + 1/(1+r) + 1/(1+r)^2 + \dots = 1/[1 - 1/(1+r)] = (1+r)/r = \$21 \text{ if } r = 5\%$$
- Oates (1969) is the classic reference on property tax capitalization
- Modern study by Cellini-Ferreira-Rothstein (2010) on school bonds in CA using regression discontinuity in vote share of local bond measures: find positive effects of bonds on house values under-investment in schools

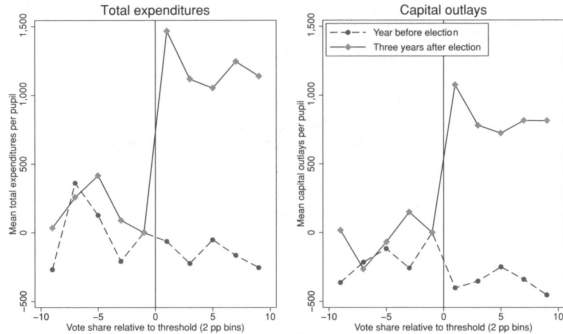


FIGURE II

Total Spending and Capital Outlays per Pupil, by Vote Share, One Year before and Three Years after Election

Graph shows average total expenditures (left panel) and capital outlays (right panel) per pupil, by the vote share in the focal bond election. Focal elections are grouped into bins two percentage points wide: measures that passed by between 0.001% and 2% are assigned to the 1 bin; those that failed by similar margins are assigned to the -1 bin. Averages are conditional on year fixed effects, and the -1 bin is normalized to zero.

Source: Cellini et al. (2010)

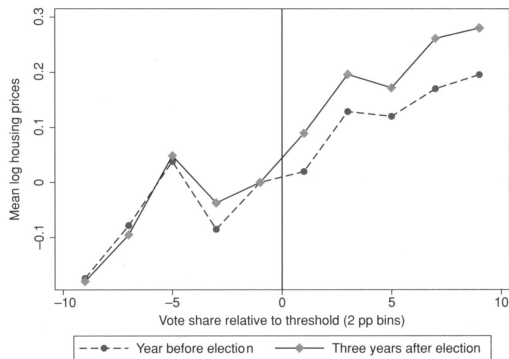
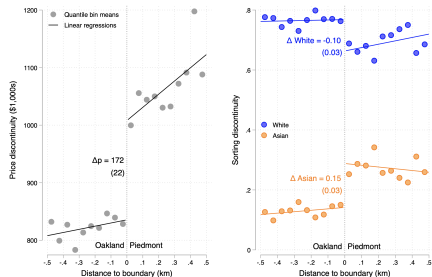


FIGURE V
Log Housing Prices by Vote Share, One Year before and Three Years after Election

Graph shows average log housing prices by the vote share in the focal bond election. Focal elections are grouped into bins two percentage points wide: measures that passed by between 0.001% and 2% are assigned to the 1 bin; those that failed by similar margins are assigned to the -1 bin. Averages are conditional on year fixed effects, and the -1 bin is normalized to zero.

Figure 8: Example: Piedmont-Oakland



Notes: Regression discontinuities in prices and racial shares (for White and Asian households) near the boundary between Oakland and Piedmont. The linear regression specification for the price discontinuity includes an indicator for being on the Piedmont side; separate linear slopes on either side of the boundary; hedonic controls (age, lot size, square feet; fixed effects for number of rooms, baths, and stories); year, race, and boundary point fixed effects.

Source: Schonholzer '23

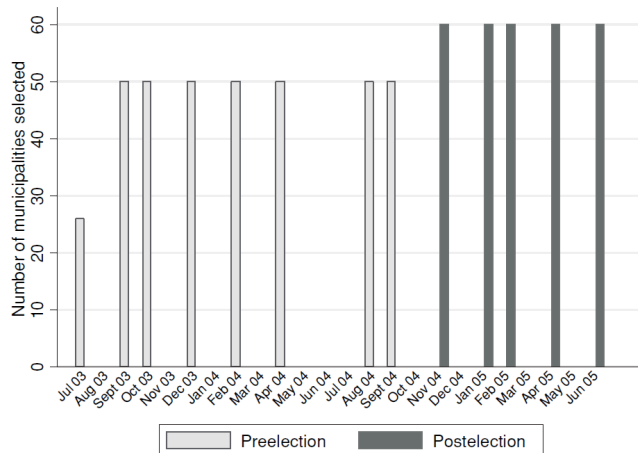
Schonholzer (2023): Analyzes the Oakland-Piedmont boundary. Piedmont is a wealthy city famed for its excellent public schools, entirely surrounded by Oakland.

Key Consequence of Tiebout Model

- It is hard for a local government to redistribute from rich to poor:
- If local redistribution is high
 - ① Poor flock to the city which provides welfare benefits
 - ② Rich flee to other cities to avoid paying for redistribution Local redistribution program will break down
- Redistribution programs work better if implemented at higher level: state or federal (harder to leave the state or country). At local level, need to have tax-benefit linkage to avoid migration
- Tax-benefit linkages: The relationship between the taxes people pay and the government goods and services they get in return.

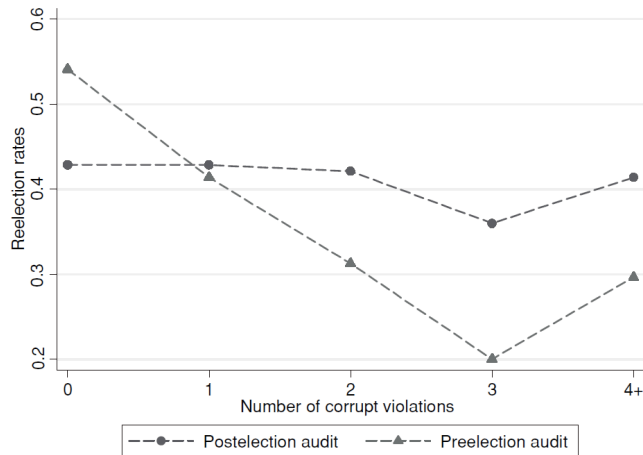
- Tiebout: Every property tax dollar efficiently goes to public service
- Major potential friction: Consumers may not know how well their property tax dollars are being spent (à la No Child Left Behind)
- Ferraz-Finan (2008): Effect of Brazilian local audits on incumbent mayors

FF: Audit release timing (treatment heterogeneity)



Source: Ferraz-Finan (2008)

Ferraz-Finan: Effect on mayors' reelection rates



Source: Ferraz-Finan (2008)

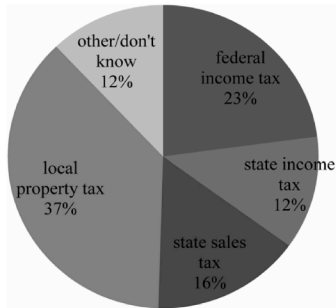
Knowledge: Do people treat it like a benefits tax?

- Tiebout: Property tax is a “benefits tax”: it pays for benefits that accrue fully to the payer, valued at cost
- Almost no other taxes are a theoretically pure benefits tax
- Should see that people think the property tax is great and other taxes are pernicious

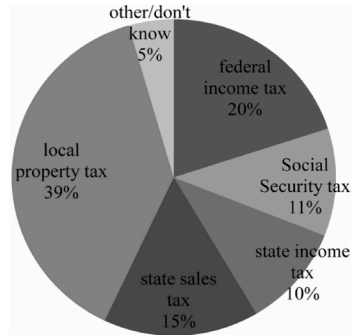
People hate the property tax

What do you think is the worst tax--that is, the least fair--federal income tax, federal Social Security tax, state income tax, state sales tax, local property tax, other/don't know?

1972



2005

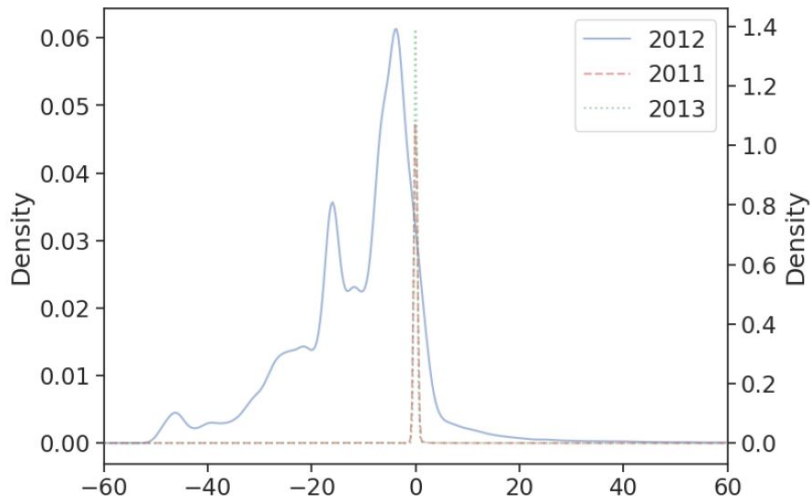


Source: Cabral and Hoxby (2013), based on polls by Advisory Commission on Intergovernmental Relations and Gallup.

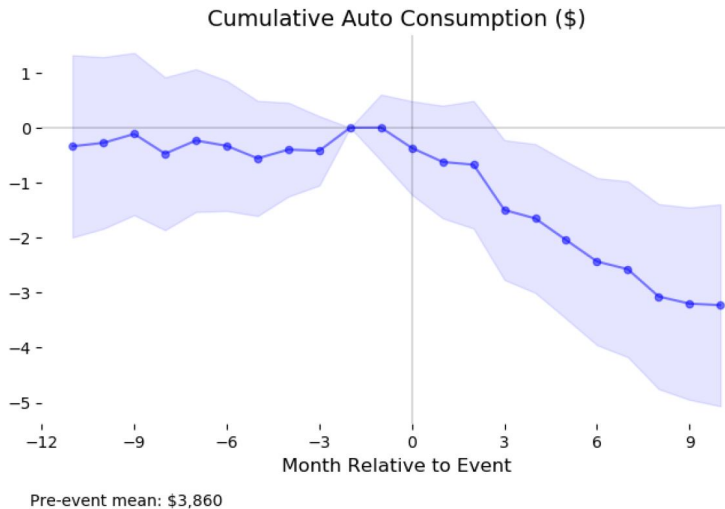
Wong (2024): Why do people hate the property tax?

- Property taxes: Assessed as fraction of home value
- Home values rise $\rightarrow$ Property tax payments rise
- Should be no liquidity problem, since can borrow against home value
- What happens when property tax bills rise?

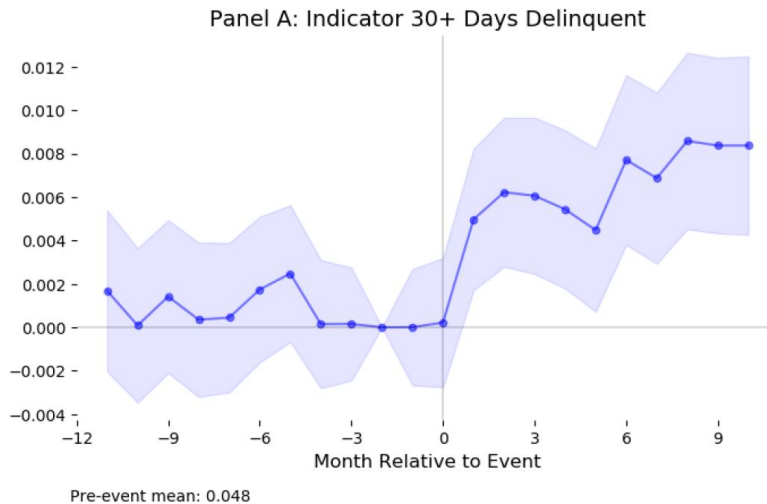
Wong (2024): Property tax reassessments



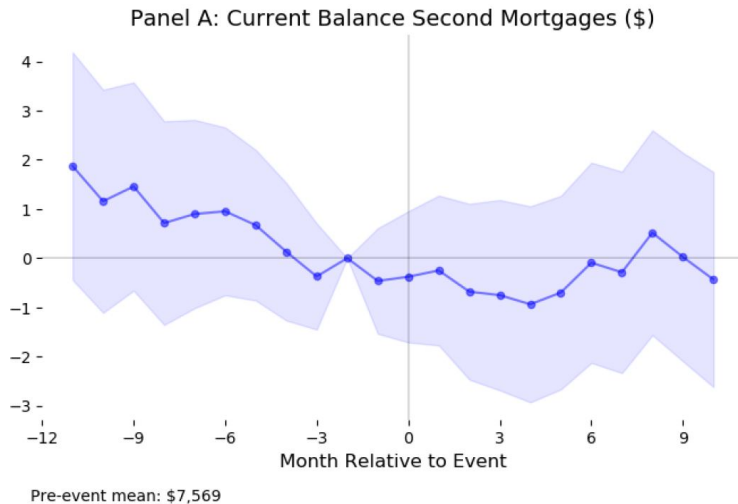
Wong (2024): Auto consumption falls ($MPC=0.27$)



Wong (2024): Mortgage delinquency rises



Wong (2024): But people don't borrow against their home



Wong (2024): People say they'd cut back on cars

Figure 8: Response to a \$500 Property Tax Increase

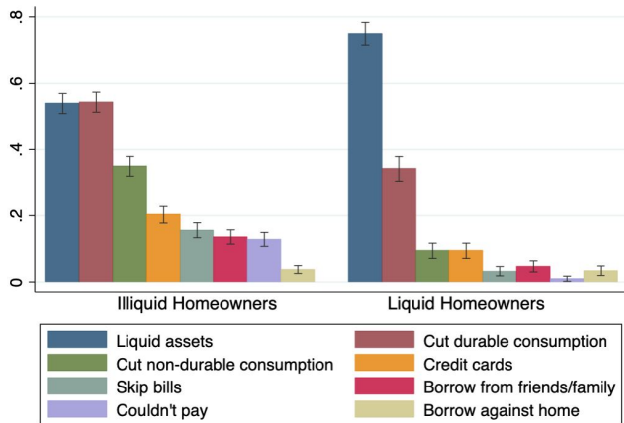
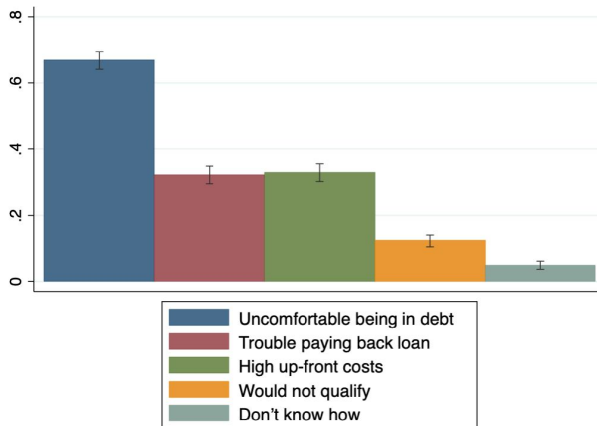


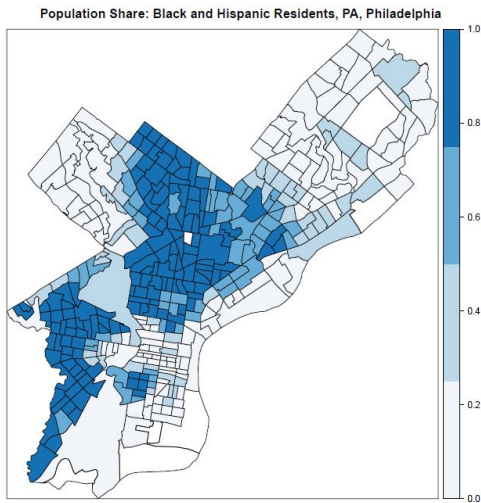
Figure 9: Reasons for Not Taking out a Second Mortgage



Wong (2024): Maybe financial hardship is key

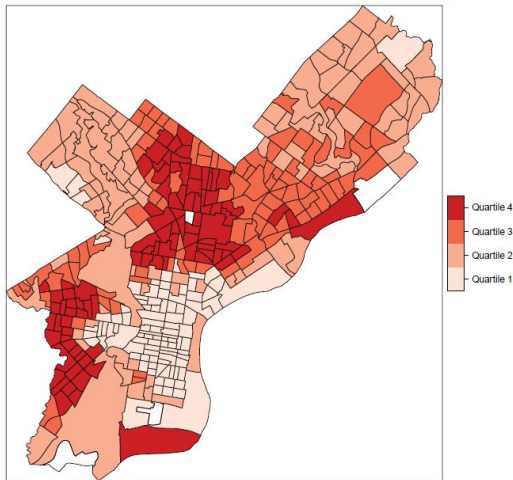
- The property tax is America's wealth tax: It does not hit a flow of cash income
- Appears to be great resistance to making people pay tax when there is no cash income (e.g., unrealized capital gains at death)
- Wong's survey: People want income-based payments
- Perhaps financial burdens of reevaluated property taxes explains unpopularity and property tax "revolts"

AvenancioLeon-Howard (2022): Black/Hispanic areas...

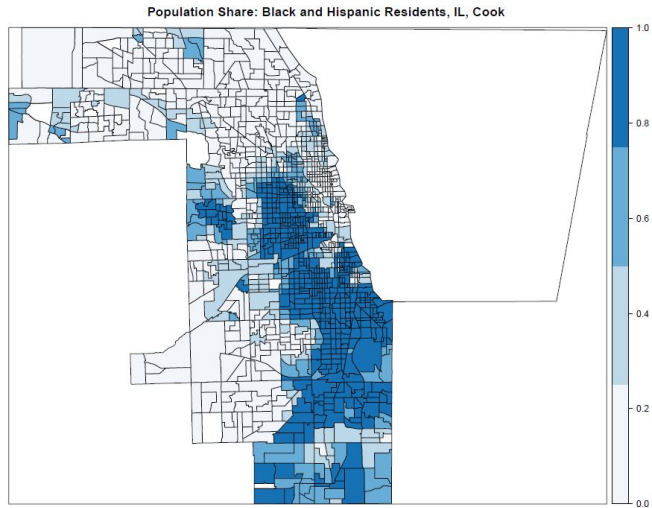


AvenancioLeon-Howard (2022): higher prop. assessments

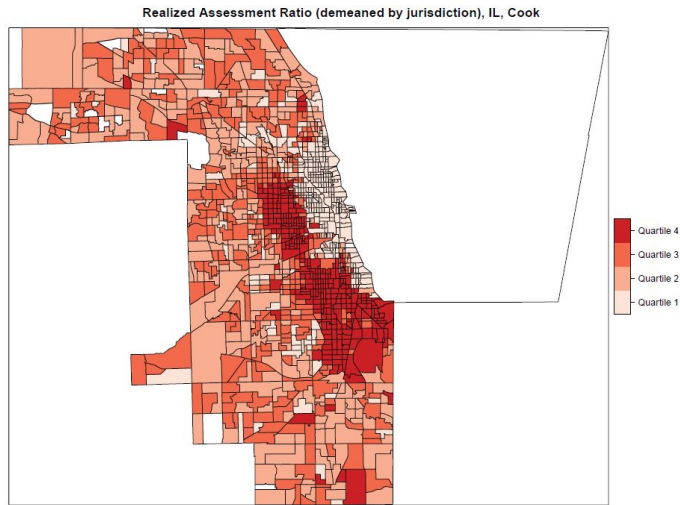
Realized Assessment Ratio (demeaned by jurisdiction), PA, Philadelphia



AvenancioLeon-Howard (2022): Black/Hispanic areas...



AvenancioLeon-Howard (2022): higher prop. assessments



AvenancioLeon-Howard (2022): Not race neutral

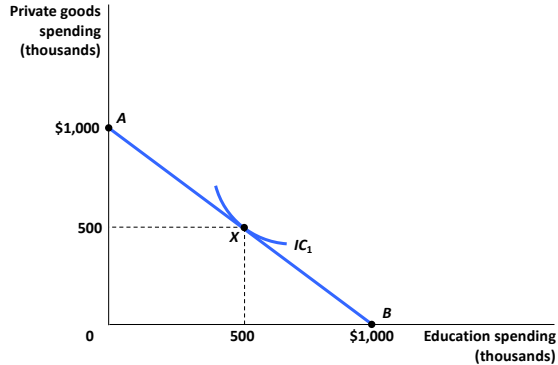
- Median Black homeowner pays \$350 extra annually, and at 90th percentile pays \$800 more annually
- Would be as if the IRS assessed higher income taxes to minorities!
- Two identifiable mechanisms
 - Too coarse of a formula
 - Whites are more likely to successfully get assessment lowered in courts
- Just using Zillow eliminates the racial gap!

Redistribution Across Communities

- There is currently enormous inequality in both the ability of local communities to finance public goods and the extent to which they do so.
- Central government can redistribute across communities directly using taxes and spending but also indirectly by giving grants to lower levels of government
- Higher levels of government can redistribute across lower levels of government through *intergovernmental grants*.
- We assume in graphical analysis that local community chooses public spending and private spending according the preferences of Median voter in the community

10.3

Tools of Redistribution: Grants

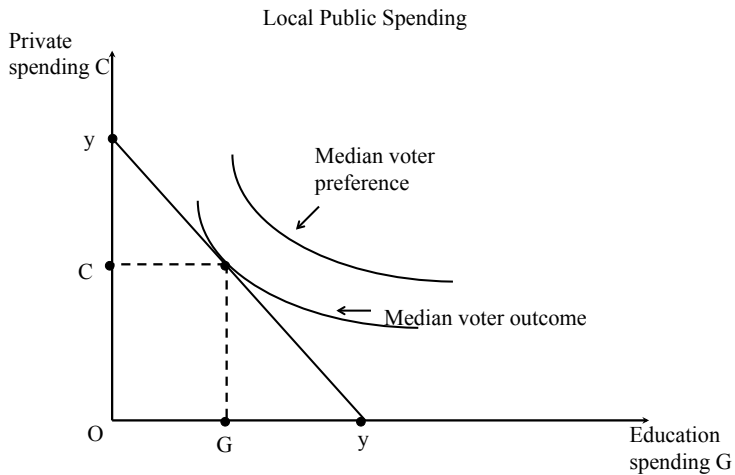


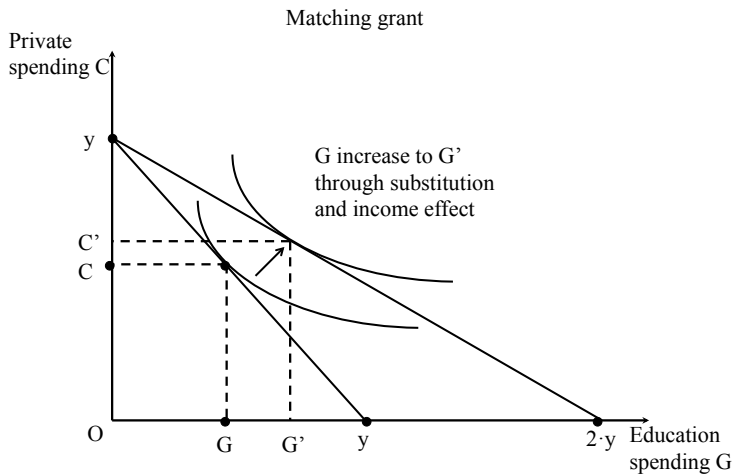
Public Finance and Public Policy Jonathan Gruber Fourth Edition Copyright © 2012 Worth Publishers

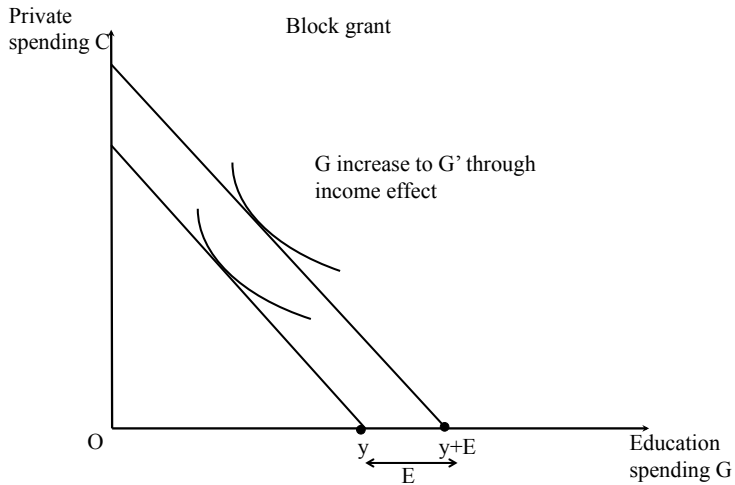
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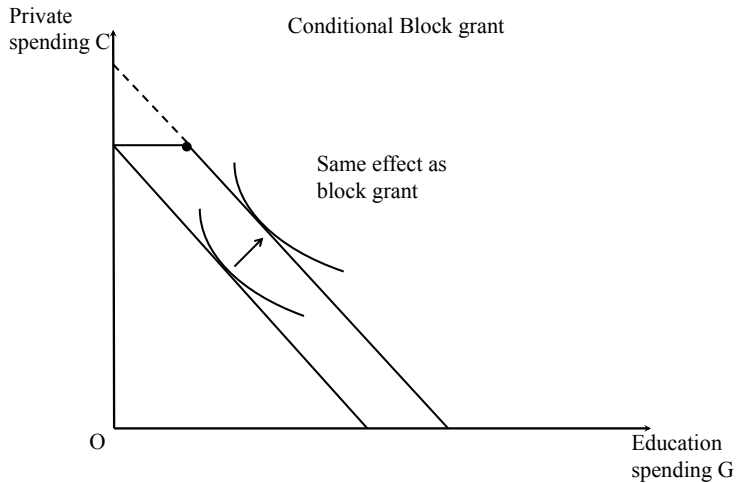
Intergovernmental Grants

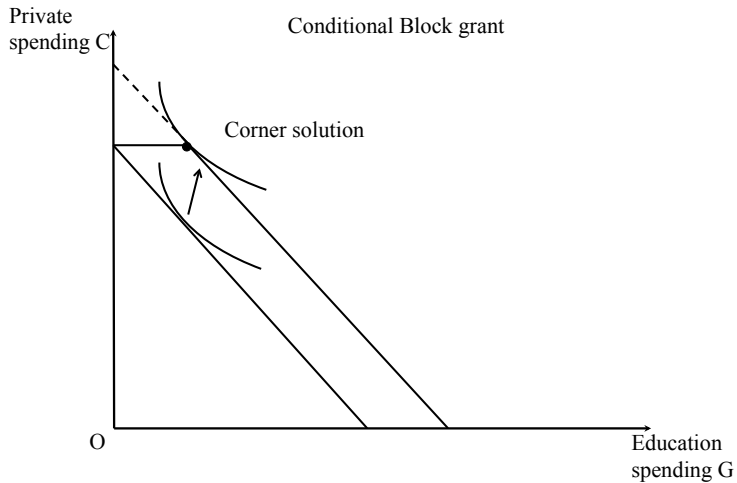
- Higher level government can provide grants to redistribute across communities and incentivize communities to spend on public goods
- Three main forms of grants:
 - ① Matching grant: A grant, the amount of which is tied to the amount of public good spending by the local community.
 - ② Block grant: A grant of some fixed amount with no mandate on how it is to be spent.
 - ③ Conditional block grant: A grant of some fixed amount with a mandate that the money be spent in a particular way.











Key Prediction of Theory: Crowd-Out

- In the theory presented, a \$1000 increase in private income has the same effect as a \$1000 increase in Fed block grant: both shift the budget in the same way and lead to the same outcome
- Example: \$1000 private income increase leads to \$800 more in private consumption and \$200 more in local taxes and public spending. \$1000 extra fed grant leads to \$200 extra in public good spending and \$800 cut in local taxes and hence \$800 extra in private consumption
- Similarly, with multiple public goods (e.g., schools and police), an extra \$1000 Fed grant for school has the same effect on schools and police than a \$1000 Fed grant for police
- Money is fungible: only total resources matter for the allocation across private good and public goods at the local level

The Flypaper Effect

- Hines and Thaler JEP'95 found that the crowd-out of state spending by federal spending is low and often close to zero
- Economist Arthur Okun described this as the flypaper effect because “the money sticks where it lands” instead of replacing state spending
- But evidence is based on correlation [not necessarily causation as states that get grants maybe the ones that like spending the most]
- Recent studies show that there is a flypaper effect in the short-run but that there is substantial crowd-out from block grants in the long-run

Redistribution in Action: School Finance Equalization

- School finance equalization: Laws that mandate redistribution of funds across communities in a state to ensure more equal financing of schools.
- Without school finance equalization, huge disparity in property tax base and hence school funding (per pupil) across areas (example from Bay Area: Lafayette is very wealthy, Richmond is poor)
- Many states (including California) impose equalization: pool local taxes at state level and redistribute them across districts
- Equalization often imposed by courts without thinking carefully about economic consequences

Redistribution in Action: School Finance Equalization

- Implicit tax on local government tax revenue: For school equalization schemes, for \$1 of extra local taxes, how much the central govt takes away in reduced transfers to local govt
 - ① With no equalization, the tax rate is 0% (local govt keeps all its revenue)
 - ② With perfect equalization, the tax rate is 100% (raising local revenue has zero impact on local spending)

California School Equalization

- In 1960s-1970s, California used to have one of the best K-12 public school systems in the nation, now it has one of the worst
- California used to have no school finance equalization and hence big disparities across areas
- 1976: Serrano vs. Priest case: California Supreme court ruled that disparities above a threshold were unconstitutional
- Wealthy districts forced to give all their tax revenue above the threshold to the common pool to fund poor districts
- local government has no incentive to raise taxes taxes and school funding fall in rich districts
- Property taxes no longer able to fund schools adequately

California Proposition 13

- In 1970s, discontent among the public about growing property taxes in CA due to:
 - ① Fast housing price increases
 - ② Local property taxes no longer funded local schools due to school equalization (prop tax not capitalized into local prices)
- Proposition 13 was voted in 1978 and imposed strong limits on property taxes (and required super majority 2/3 vote in state legislature to increase ANY tax):
 - Assessed value of real estate property can only grow at most by 2% per year (instead of following price increases which are around 4-5% on average)
 - Property owners no longer face big increases in prop tax (helps retirees on fixed income)
 - New owners end up paying much more than old owners (e.g., house assessed at \$200K that sells for \$1m will see a 5-fold increase in property taxes). Creates a lock-in effect (Ferreira 2010)

Poll (No Right Answer)

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- A. True
- B. False
- C. Uncertain

References

- Jonathan Gruber, Public Finance and Public Policy, Fifth Edition, 2016 Worth Publishers, Chapter 10
- Avenancio-León, Carlos F., and Troup Howard. 2022. “The Assessment Gap: Racial Inequalities in Property Taxation.” The Quarterly Journal of Economics 137(3), 1383-1434. [\[web link\]](#)
- Cabral, Marika, and Caroline M. Hoxby. 2012. “The Hated Property Tax: Salience, Tax Rates, and Tax Revolts.” NBER Working Paper 18514. [\[web link\]](#)
- Cellini, Stephanie, Fernando Ferreira and Jesse Rothstein. 2010. “The Value of School Facility Investments: Evidence from a Dynamic Regression Discontinuity Design”, Quarterly Journal of Economics, 125(1), 215-261. [\[web link\]](#)
- Ferraz, Claudio, and Frederico Finan. 2008. “Exposing Corrupt Politicians: The Effects of Brazil’s Publicly Released Audits on Electoral Outcomes.” The Quarterly Journal of Economics 123(2), 703-745. [\[web link\]](#)

References

- Ferreira, Fernando. 2010. "You can take it with you: Proposition 13 tax benefits, residential mobility, and willingness to pay for housing amenities." *Journal of Public Economics* 94(9-10), 661-673. [\[web link\]](#)
- Hines, James R., and Richard H. Thaler. "Anomalies: The flypaper effect." *The Journal of Economic Perspectives* 9.4 (1995): 217-226. [\[web link\]](#)
- Hoxby, Caroline M. "Does Competition among Public Schools Benefit Students and Taxpayers?." *The American Economic Review* 90.5 (2000): 1209-1238. [\[web link\]](#)
- Oates, Wallace E. 1969. "The Effects of Property Taxes and Local Public Spending on Property Values: An Empirical Study of Tax Capitalization and the Tiebout Hypothesis" *Journal of Political Economy* 77(6), 957-971. [\[web link\]](#)
- Rothstein, Jesse. "Does Competition Among Public Schools Benefit Students and Taxpayers? Comment." *The American Economic Review* 97.5 (2007): 2026-2037. [\[web link\]](#)

- Monarrez, Tomás E., and David Schönholzer. 2023. “Dividing Lines: Racial Segregation across Local Government Boundaries.” *Journal of Economic Literature* 61(3), 863-887. [\[web link\]](#)
- Tiebout, Charles M. “A pure theory of local expenditures.” *The Journal of Political Economy* 64.5 (1956): 416-424.[\[web link\]](#)
- Wong, Francis. “Taxing Homeowners Who Won’t Borrow.” (2024). [\[web link\]](#)